

Keyboarding Techniques

HOW OFTEN do you type using a keyboard? You probably type when writing emails or papers on a personal computer. You probably type when chatting in games on your personal computer. You will probably continue typing for most of your life. You will likely need to type during your adult career. This lesson focuses on keyboard technique, speed, and accuracy.



Objective:



Demonstrate effective keyboarding techniques and identify ways to increase keyboarding speed and accuracy.

Key Terms:



modifier (combination)
keys
function key
gross words a minute
(GWAM)

hand position
home row
net words a minute
(NWAM)
numeric keypad

posture
QWERTY keyboard
shift key

Keyboarding Techniques

KEYBOARD TERMINOLOGY

A **QWERTY keyboard** refers to the common layout of keys on a personal computer or laptop keyboard, getting its name from the sequence of letters in the upper left corner of the alphabet on the keyboard. QWERTY keyboards have been around since the rise of popularity in typewriter in the 1780s. Since that time, the QWERTY keyboard layout has changed with technical advances; however, the placement of letters has remained the same over time. Some

additional keys have been added to the keyboard, accompanying the letters and improving how you can work with a document while keyboarding. This section summarizes types of keys often found on a contemporary personal computer or laptop QWERTY keyboard.

Alphabet and Home Row

Alphabet keys are the keyboard letters used to type the text content of a document. The **home row** refers to the letters that are the starting and resting keys for your fingers. Your left index finger rests on the letter F, and your right index finger rests on the letter J. Both letters F and J have bumps on them so they can be found easier without looking at the keyboard. On the home row your left-hand fingers rest on ASDF and right-hand finger rest on JKL:.

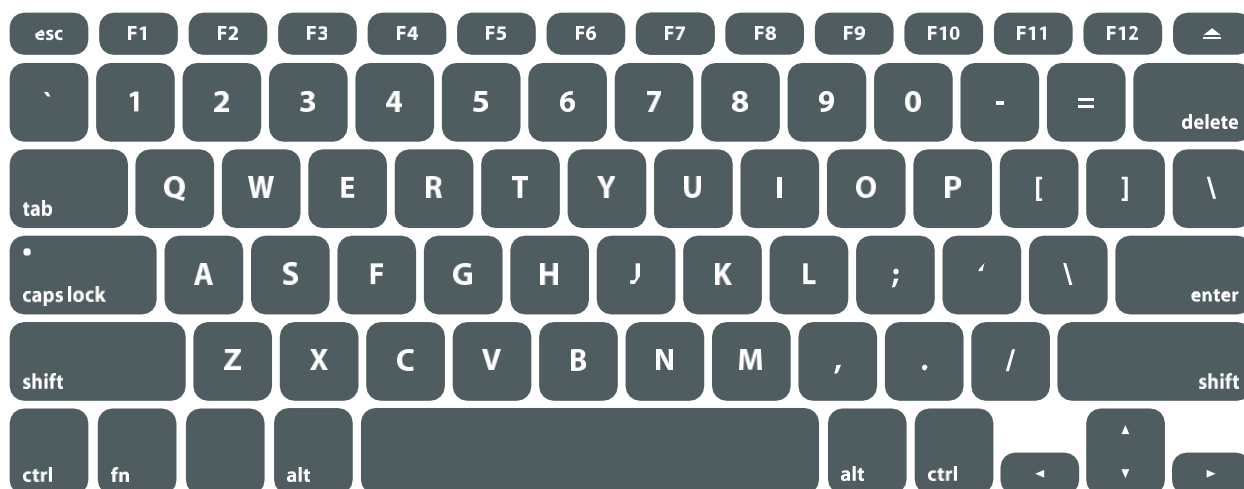


FIGURE 1. The typical layout of a QWERTY keyboard.

Modifier (Combination) Keys

Modifier (combination) keys are special keys that changes what another key does when the two keys are pressed together. The control key is typically labeled “ctrl” and is located on both sides of space bar on the bottom row. The control key “ctrl” can be pressed with the letter I to italicize text; can be pressed with the letter U to underline text; and can be pressed with the letter B to bold-type text. Another example is the “shift” key, which is located on both sides of the bottom alphabet row. The **shift key** performs the task of capitalizing a letter when pressed together.

The function key is typically labeled “fn” and is often found on the lower left near the space bar. The **function key** modifies the task other “f” keys perform when the two keys are pressed at the same time. When the “fn” key is pressed in combination with “f1, f2, f3... keys” at the top of the keyboard, it causes the computer to perform certain tasks. For example, you can control the brightness of screen or volume using “f” keys. The job of “f” keys in the top row will be different for different computers; and the f keys on the top row sometimes have an icon on them to help you see what they do. These keys work in combination with the “fn” key in the bottom row though.

The alternate key is typically labeled “alt” and is located on the lower row on both sides of the space bar. The alt key modifies the task of one or more keys when they are pressed at the same time. When the alt key is pressed with ctrl and delete keys, the computer task manager is brought up. This is often used if a program has frozen up the computer and needs to be shut down to free the computer back up to work properly. You would press ctrl+alt+delete and the task manager will appear with a list of programs running. You select the program causing the problem and ask the task manager to shut that program off, leaving your other programs working. If an Internet website is freezing your computer during research writing, you can close the problematic website browser this way without losing your written document.

Modifier (combination) keys offer more agility and functionality to the QWERTY keyboard. Many of the modifier (combination) options are not used during a basic typing activity. However, some of the options such as using shift for capitalization are used all the time. It is recommended that you learn how these modifier (combination) keys work on your keyboard. Do your learning and experimenting during free time and not while working on an important assignment. Experimenting with these keys in the middle of an important assignment is not recommended because it could disrupt your work or might cause you to lose control of your keyboard. Learn the modifier (combination) keys that are most helpful, and use them to make your work easier while typing an assignment or important document.

Numeric Keypad

Most QWERTY keyboards also have a numeric keypad on the far-right side. Data entry tasks require the use of this numeric keypad most often. When entering numeric data, you will use your right hand to type on the numeric pad, with the middle finger using the number 5 as the home row. The number 5 will have a bump on it to help find it without looking. From there, the first three fingers (not the pinky) is used to type numeric data. The thumb is used to type zero at the bottom of the pad.



FIGURE 2. Example of a QWERTY keyboard with numeric pad.

IMPORTANT KEYBOARDING TECHNIQUES

When you sit down to type, there are important keyboarding techniques to follow which will help improve your ability to type. This section explains basic techniques to follow when keyboarding. These points focus on keeping the appropriate body posture and hand position. **Posture** refers to the best position of your body in relation to the keyboard. **Hand position** refers to the best method for placing and keeping your hands interacting with the keyboard.

Posture

The first technique is to sit up straight on a chair or stool, with the keyboard in front of you on a flat surface so that you are able to reach the keyboard effortlessly at about the level of your stomach. The second technique is to sit directly in front of the keyboard, facing it with your back as straight as possible and both feet on the ground in front of you, usually slightly under the surface the keyboard is sitting on. Your upper legs and feet should be parallel to the floor. The keyboard monitor should be at eye-level and preferably with a slight tilt.

Hand Position

The third technique is that your arms should be relaxed at your sides, with your fingers on the home row, wrists either hovering above the keyboard or resting on a pad made specifically for keyboard wrist resting. The fourth technique is that your fingers should be comfortably



ON THE JOB...

CAREER CONNECTION: Typing on the Job

Plenty of jobs require keyboarding/typing. Some common job titles that require typing at a minimum number of words per minute are as follows: secretary, receptionist, legal or administrative assistant, data entry, and court reporter. These positions typically require between 60 to 100 words per minute. Court reporters are expected to type much faster, upwards of 200 words per minute. Some other positions such as Certified Nurse, chefs, cosmetologists, or sales people may not type often and may only need to type 25 to 40 words per minute. Whatever the case, one is better prepared and efficient with completing keyboarding tasks by learning, practicing, and using the most effective keyboarding techniques.



curved and able to easily rest on the home row, while also being able to move as needed to all of the keys on the keyboard. Both thumbs should be able to rest on the space bar at the bottom of the keyboard.

CORRECT SITTING POSTURE

COMPUTER WORKSTATION ERGONOMICS

MONITOR

Adjust distance and height: top of the monitor at eye-level and slightly tilted.

18/24 in (45-70 cm)

ARMS

Relax shoulders, forearms parallel to the floor. Minimal bend at the wrist.

CHAIR

Should have a backrest and armrests, adjust height.

LEGS

Thighs parallel to the floor.

FEET

Parallel to the floor, use a footrest if necessary.



TAKE BREAKS
EVERY 25 MINS

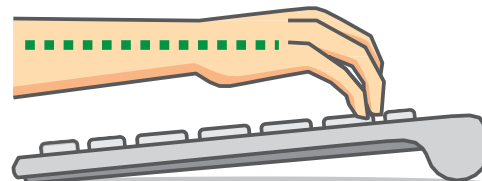
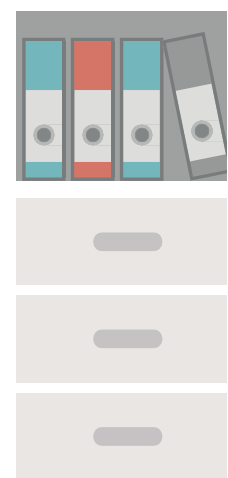


FIGURE 3. Proper posture and hand positioning at the keyboard.

TECHNIQUES FOR INCREASING KEYBOARDING SPEED AND ACCURACY

There are two ways of measuring typing speed and accuracy. **Gross words a minute (GWAM)** is the total number of words typed per minute including errors. **Net words a minute (NWAM)** is the total number of typed words per minute minus errors. There are three elements to improving speed and accuracy. First, using the appropriate posture and hand positioning is required. Second, memorization of keys and finger assignment is needed. Third, using effective practice methods is needed.

Finger Assignment

To begin typing, place both hands on the home row. From there, each finger is assigned to reach specific keys. Your hands stay positioned near the home row, while each finger reaches to its assigned keys. Finger assignment and reach must be memorized and practiced without looking at the keyboard. When typing, one usually looks at the text one is typing and does not look at the keyboard directly. The typing is done from memory of finger assignment. Finger key assignment is summarized in FIGURE 4.

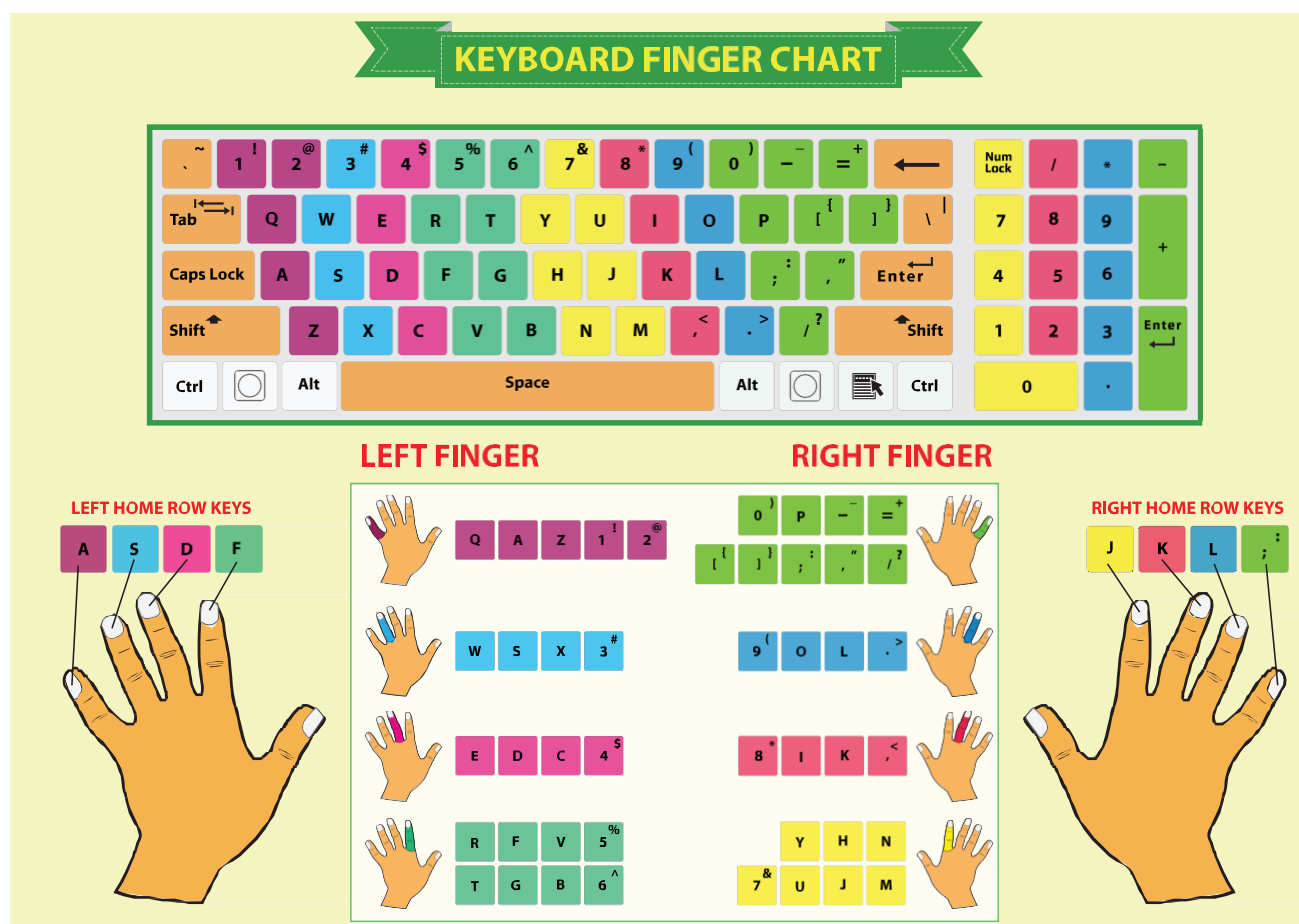


FIGURE 4. Finger assignments.

Practice Methods

There are different programs that help learn to type. Schools offer typing class. Typing classes are offered online as well. Practice methods normally help memorize the keys and finger assignment at first. Learning the keys and finger placement comes first. Second, more complicated sentences are used to practice, to help you learn to reach each key accurately. When practicing, gross words a minute and net words a minute are both measured. With practice and taking typing tests, your gross words a minute will increase and your errors will decrease.

Following is a famous sentence that can be used to practice every letter of the alphabet. The quick brown fox jumps over the lazy dog. If you practice typing this sentence, it will enable you to reach every letter of the alphabet. When trying to increase speed, practice is faster with more errors. When practicing accuracy, speed is normally slower. As you practice longer and consistently, accuracy increases at the same time speed increases.

Summary:



QWERTY refers to the layout of keys on a keyboard, getting its name from the sequence of letters in the upper left corner of the alphabet on the keyboard. The home row refers to the letters that are the starting and resting keys for your fingers. On the home row your left-hand fingers rest on ASDF and right-hand finger rest on JKL:. Modifier (combination keys) are special keys that change what another key does when the two keys are pressed together. Control, function, shift, and alt are examples of modifier keys. Posture and hand position are important techniques for learning to type. Typing speed and accuracy are measured using gross words a minute and net words a minute. There are three elements to improving speed and accuracy. First, using the appropriate posture and hand positioning is required. Second, memorization of keys and finger assignment is needed. Third, using effective practice methods is needed.

Checking Your Knowledge:



1. Which letters are the home row for left and right hands?
2. Name three examples of modifier (combination) keys.
3. What keys are assigned to the right index finger?
4. What does 'gross words a minute' and 'net words a minute' measure?
5. What famous sentence is used to practice typing every letter of the alphabet?

Expanding Your Knowledge:



Complete an Internet search to identify free online typing practice resources. One example is Typing Club at <https://www.typingclub.com/>. What are other resources you might use to practice keyboarding/typing?

Web Links:



Best Typing Software

<https://www.toptenreviews.com/best-typing-software>

Keyboarding Tips and Information

<http://keyboarding.ccsd.edu/help-for-students/typing-position>